



CCS 2025
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CLIMATE-INDUCED DELAY PROPAGATION IN RAIL NETWORKS AS A NON-MARKOVIAN PROCESS

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COMPLEX MULTILAYER NETWORKS

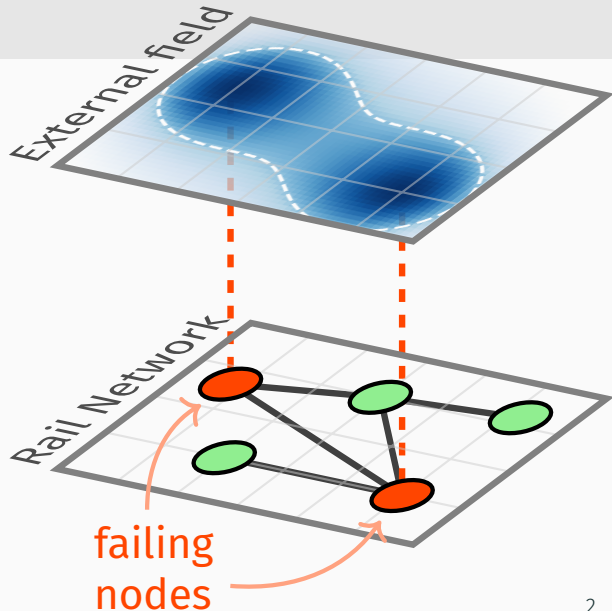
Framework

3 ingredients for this soup:

- Infrastructure: complex topology embedded in space
- Dynamical model
- External climate stressors

Research Question

Interaction between human-made systems and natural external stressors.



Modelling the system perturbations

Non-Markovian *propagation*
model with *generation* and
absorption processes

$$\mathbf{s}(\mathbf{t} + \mathbf{1}) = \mathcal{T}(\mathbf{t})\mathbf{s}(\mathbf{t}) + \beta\mathbf{g} - \gamma\mathbf{c}$$

$\mathbf{s}(\mathbf{t}) \rightarrow$ *perturbation state* vector
defined at each station (delay)

Real world application:

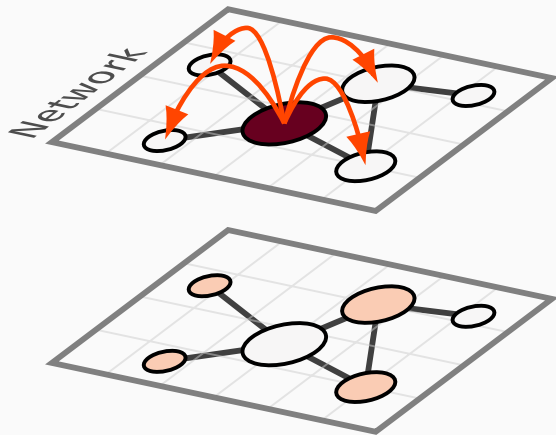
-  *Delay propagation on a*
-  *railway system influenced by*
-  *rainfalls*

Model — Diffusion process

The diffusion process ruled by the **network topology**

$$\mathbf{s}(\mathbf{t} + 1) = \mathcal{T}(\mathbf{t})\mathbf{s}(\mathbf{t}) + \beta\mathbf{g} - \gamma\mathbf{c}$$

$\mathcal{T}(\mathbf{t})$ transfer probability



Model — Generation process

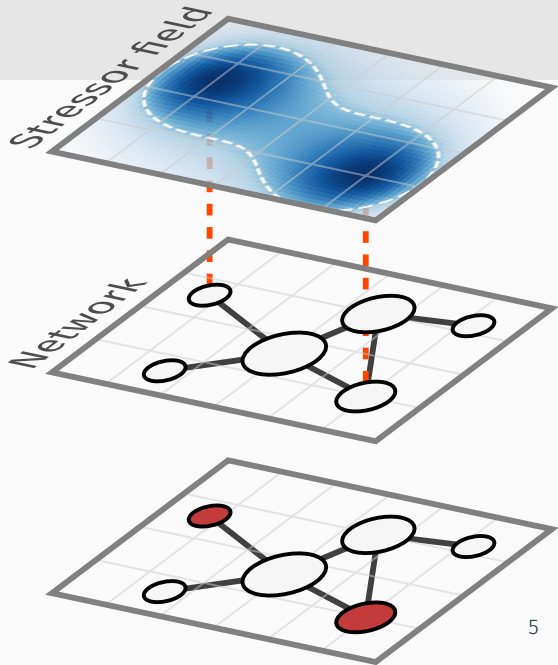
The generation process ruled by the **external stressor field** (rain)

$$\mathbf{s}(\mathbf{t} + 1) = \mathcal{T}(\mathbf{t})\mathbf{s}(\mathbf{t}) + \beta \mathbf{g} - \gamma \mathbf{c}$$
$$\mathbf{g} = (\mathcal{W}(\mathbf{t}) \odot \mathcal{S}(\mathbf{t}))\mathbf{1}$$

β generation rate

$\mathcal{W}(\mathbf{t})$ trains traversing each link

$\mathcal{S}(\mathbf{t})$ amount of rain experienced at each link



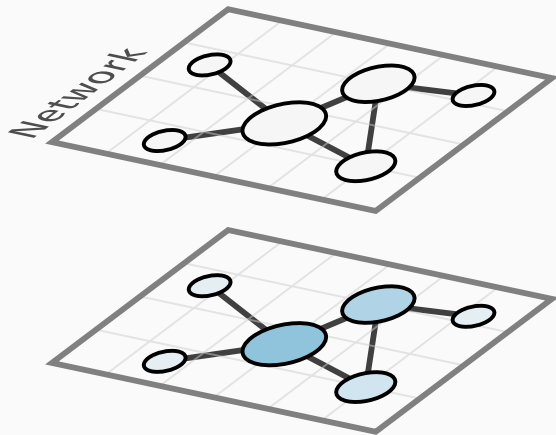
Model — Absorption process

The absorption process ruled by the **local capacity** to absorb delays

$$\mathbf{s}(\mathbf{t} + \mathbf{1}) = \mathcal{T}(\mathbf{t})\mathbf{s}(\mathbf{t}) + \beta\mathbf{g} - \gamma\mathbf{c}$$

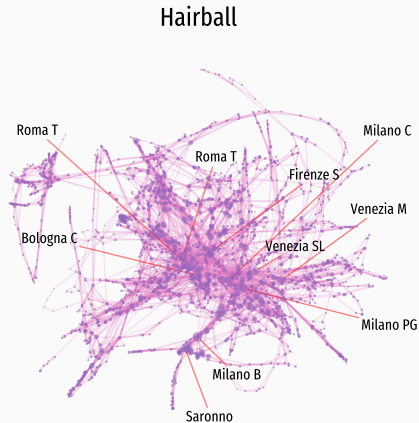
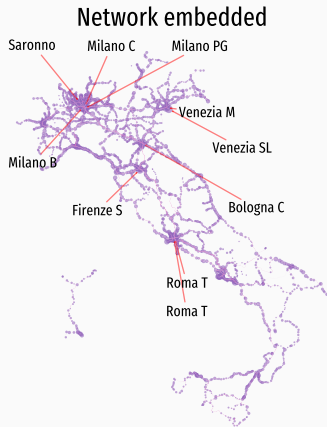
γ absorption rate

$\mathbf{c}$ station capacity





- Network of stations connected by direct trains.
- Shapes and locations from OpenStreetMap

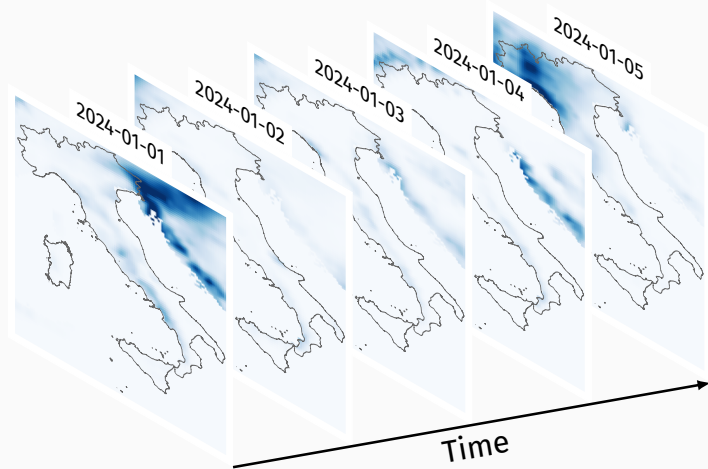


Not a Ridiculogram since not published in *Nature* nor *Science*

Data — Stressor

Historical weather variables
(*hourly*; ERA5@Copernicus)

`total_precipitation_land`



Data — Real world

- All trains (2020-2024) from the major Italian railway operator
- Arrival/departure scheduled and real time

DEPARTURES

DESTINAZIONE
DESTINATION

ORARIO
TIME

RITARDO
DELAY

INFORMAZ
INFORMAT

TORINO P.N.

13:45

200'

RMA

BOLZANO

16:15

55'

RMA

MILANO C.LE

16:20

55'

RMA

MILANO C.LE

16:30

45'

RMA

BOLZANO

16:45

35'

RMA

TORINO P.N.

17:00

15'

RMA

FIUMICINO A.

17:05

RMA

LATINA

17:06

RMA

CASSINO

17:07

RMA

MILANO C.LE

17:10

15'

RMA

FIRENZE SMN

17:12

RMA

GROSSETO

17:12

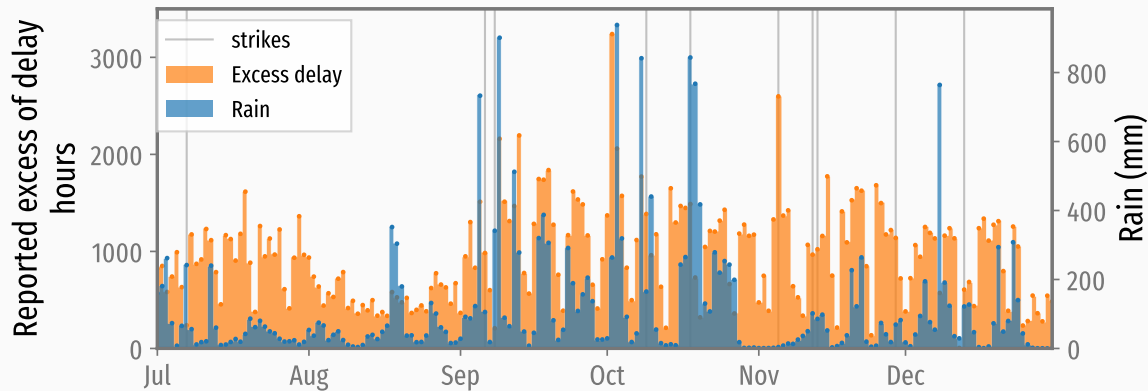
RMA

FROSINONE

17:14

RMA

Delays originate from multiple sources



Cumulated daily delay vs. rainfalls → multiple sources of delay

Results

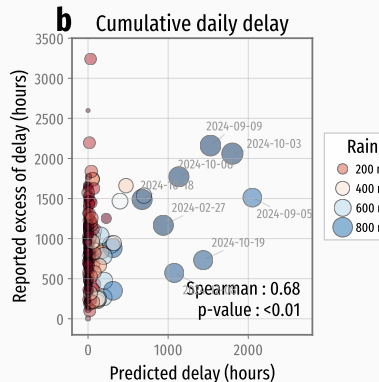
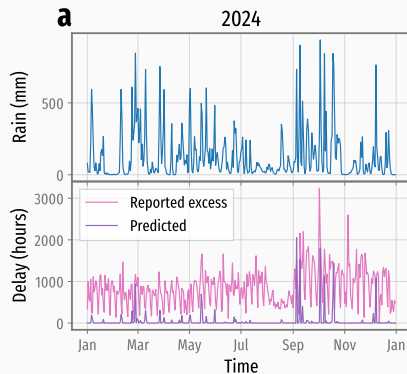
Parameter inference

- Learning set: 2021-2023.

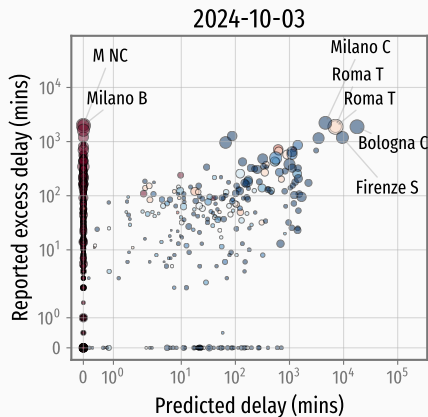
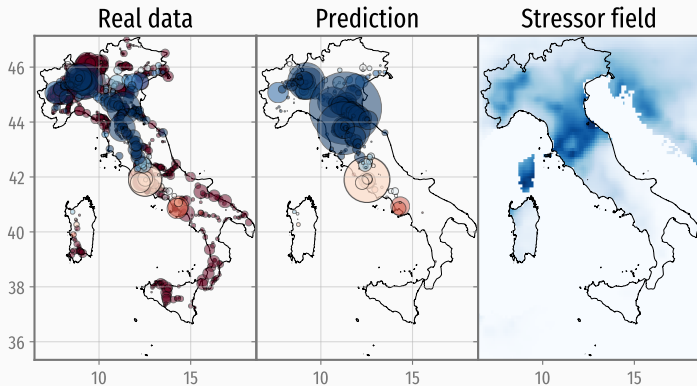
- Validation: 2024

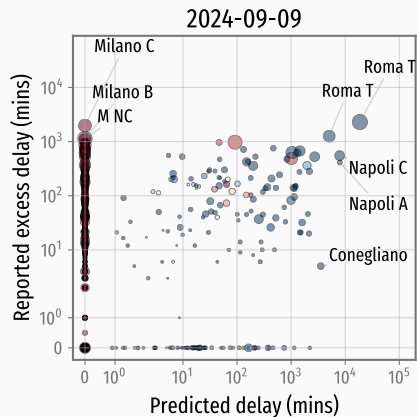
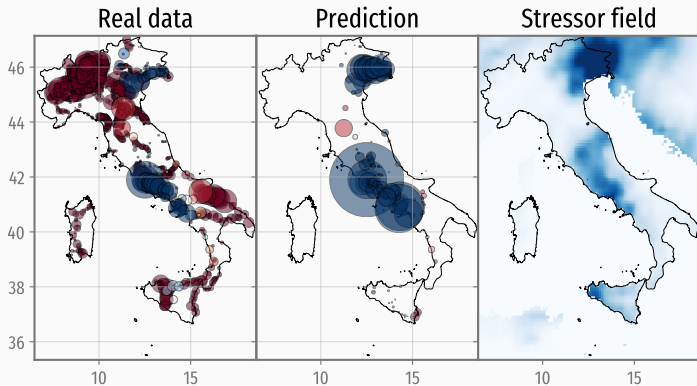
We can predict delay distribution in rainy (blue) days.

Spearman: 0.68

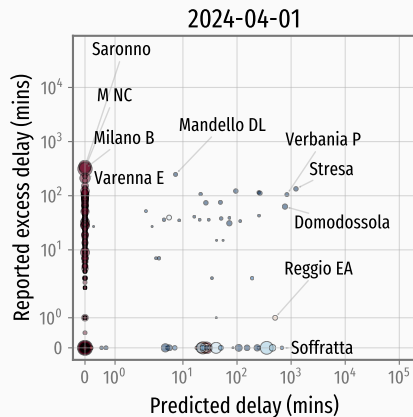
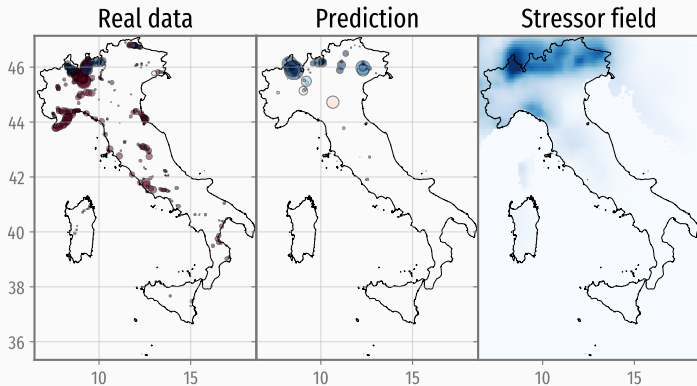


2024-10-03





2024-04-01



Phase space — System resilience

System response from a continuous external stressor field.

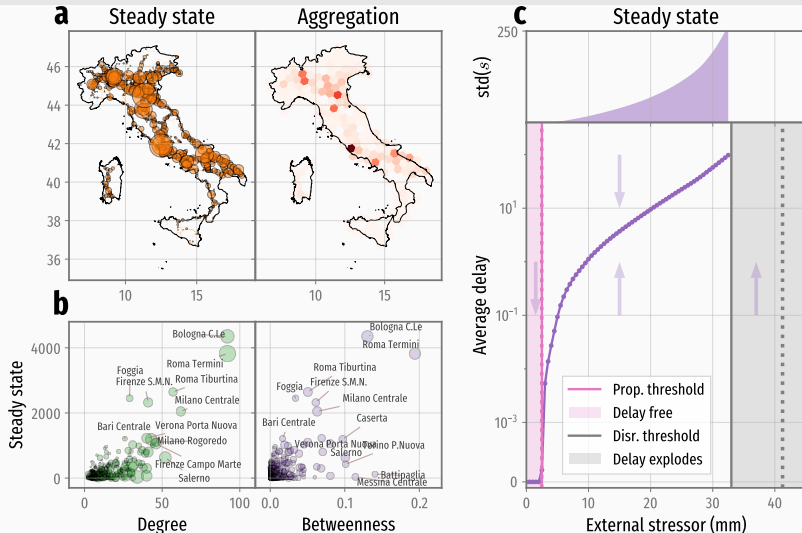
- *Delay absorbed below a propagation threshold*

$$\beta g_i < \gamma c_i \quad \forall i$$

- *System collapse above a disruption threshold*

$$\langle \beta g \rangle > \langle \gamma c \rangle$$

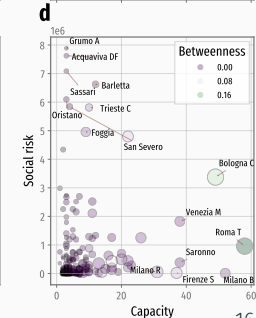
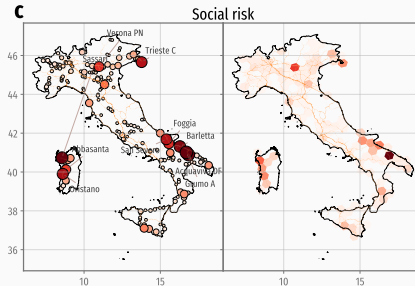
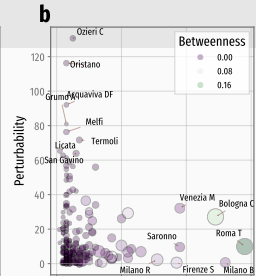
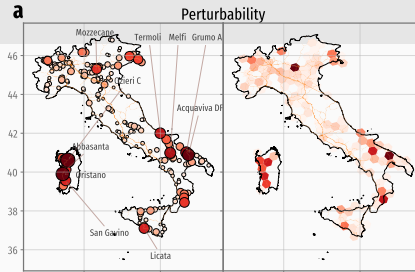
- *Endemic delay in-between (dynamical equilibrium)*



Perturbability and Social risk

Perturbability: station ability to influence the network
Total delay experienced by the network after initial stress around a station.

Social risk: perturbability weighted by the *social cost* (population affected)



? Questions?

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Joint work with:



Tomas Scagliarini



Manlio De Domenico



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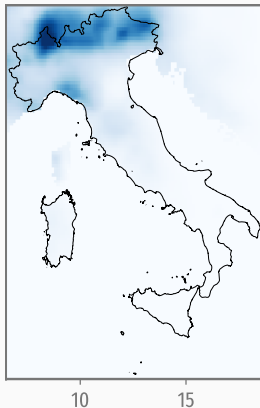
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COMPLEX MULTILAYER NETWORKS

- Model of *propagation+generation+absorption* of perturbations
- Interaction between external stressor (e.g. climate variables) and dynamics
- Prediction of *complete system failure* at high values of the external stressor
- *Perturbability centrality* and *social risk*

Aknowledgements: FIS project no. FIS00000158 (CUP C53C23000660001). INFN (LINCOLN).

Perturbability and Social risk

Stressor field



What happened in
2024-04-01 🐟?

Stressor field *perpendicular* to
the *system perturbability*.

